Fundamental Concepts

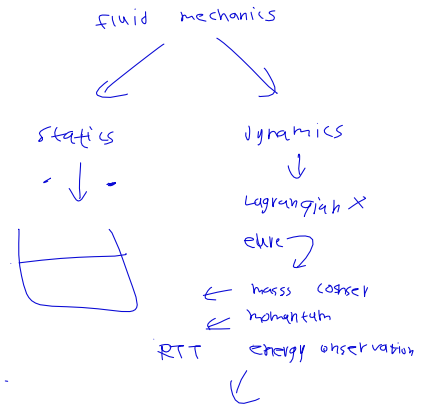
CE2201 Fundamentals of Fluid Mechanics

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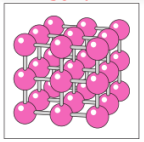
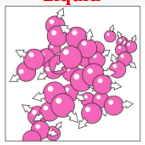
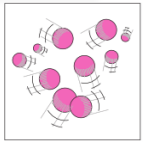
E2024 Batch-26th- Year 2025/2026

Introduction

- Fluid mechanics deals with the study of behavior of **fluids at rest (*fluid statics)*** or **in motion (*fluid dynamics)***, *and the interaction of fluids with solids* or other fluids at the boundaries.
- Fluid is a substance in the liquid or gas phase
- Distinction between a solid and a fluid

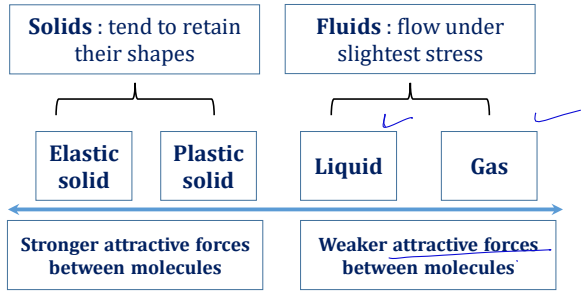


Distinction between a solid and a fluid

Solid	Liquid	Gas
		
• Molecules are at relatively fixed positions	• Groups of molecules move about each other	• Molecules move about at random
	• Volume remains relatively constant due to strong cohesive forces	• Cannot form a free surface. Fills the entire available space
	• A liquid takes the shape of the container it is in, and it forms a free surface	• Gas molecules are widely spaced, and the cohesive forces between them are very small
Molecular attraction $F = \frac{Gm_1m_2}{r^2}$ Small r -> large F		
Stronger attractive forces between molecules		Weaker attractive forces between molecules

Distinction between a solid and a fluid

The strength of intermolecular cohesive forces determines the status of a particular type of material



Normal stress and shear stress

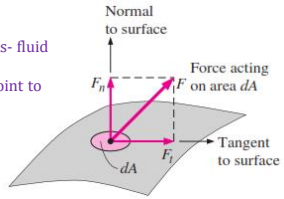
Stress= Force/Area

Normal stress

$(\sigma) = \frac{F_n}{dA}$

Normal compressive stress- fluid pressure

p= p(x,y,z,t) varies from point to point



Tangential stress (τ) = $\frac{F_t}{dA}$

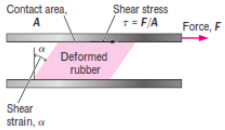
Shear stress: Friction due to fluid viscosity is the primary source of shear stresses in a fluid.

For fluids at rest, the shear stress is zero and pressure is the only normal stress

Distinction between a solid and a fluid

Solid

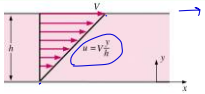
- resist an applied shear stress by deforming
- stress is proportional to *strain*



Deformation of a rubber eraser placed between two parallel plates

Fluid

- deforms continuously under the influence of shear stress
- stress is proportional to *strain rate*



the fluid layer in contact with the upper plate would move with the plate continuously at the velocity of the plate

Fluid properties

- The state of a system is described by its properties.
- Intensive properties:

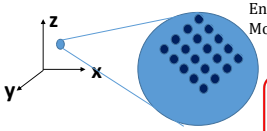
  Independent of the mass of the system (Temperature, pressure, density)

- Extensive properties:

  Depends on the size or extent of the system (mass, volume, momentum)

Fluid properties

- Continuum: view it as a **continuous, homogeneous matter with no holes**, that is, a **continuum**. The continuum idealization allows us to treat properties as point functions and to assume that the properties vary continually in space with no jump discontinuities.



Enlarged view
Molecular structure of matter

Molecular diameter 10^{-6} mm
Separated by 10^{-6} mm~10^{-7} mm
In a small volume of 1 mm^3 approximately 10^{20} molecules

In practical applications we consider CV much higher with very large no of molecules. Ignore voids

Fluid density

- Mass per unit volume

  $\rho = \frac{m}{v}$ Unit: kg/m^3

- The density, in general, depends on temperature and pressure. The density of liquids and solids depends more strongly on temperature than it does on pressure

- Specific weight (γ)- Weight of unit volume

  $\gamma = \rho g$

$$F = \frac{\mu A (V_2 - V_1)}{d}$$

$V_1 = 0$

Q1) Two parallel walls spaced at 25 mm is filled with a liquid of dynamic viscosity 0.7 Pa s. A thin flat plate of 250 mm x 250 mm is towed into this space at a velocity of 150 mm/s at a distance of 6 mm from one wall keeping the movement parallel to the walls. Assuming linear velocity variations between the plate and the walls, determine the shear force exerted by the liquid on the plate.

$$F = \frac{\mu A V}{d = h}$$

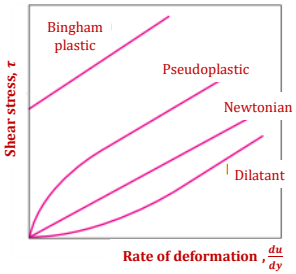
Fluid viscosity

- When a fluid moves relative to a solid or when two fluids move relative to each other a friction force develops at the contact surface in the direction opposite to motion.
- Property that represents the internal resistance of a fluid to motion or the "fluidity," and that property is the **viscosity.**
- **Newtonian fluids:**

  $\tau = \mu \frac{du}{dy}$

- Kinemetic viscosity:

  $\nu = \frac{\mu}{\rho}$ Unit m^2/s



$\tau = \mu \frac{du}{dy}$

Newtonian fluids

kinematic viscosity ⇒ $\nu = \frac{\mu}{\rho}$

Fluid compressibility

- Volume (or density) of a fluid changes with a change in its temperature or pressure.
- Fluids usually expand as they are heated or depressurized and contract as they are cooled or pressurized.
- Coefficient of compressibility (1/K) and Bulk modulus (K)

  $K = -\frac{\Delta P}{\Delta V / V}$ In liquids spacing is smaller than gas

  $K_{water} \sim 2.25 \times 10^9\ Pa$; $K_{air} \sim 1.4 \times 10^5\ Pa$

- K is very large for liquids and small for gasses. For a given pressure change,

  Liquids: volume change and density change is negligible for liquids. → V, ρ

  constant density fluids or incompressible fluids

  Gases: Density change is considerable for gasses →

  variable density or compressible fluids.

Coefficient of Compressibility $= \left(\frac{1}{k}\right)$

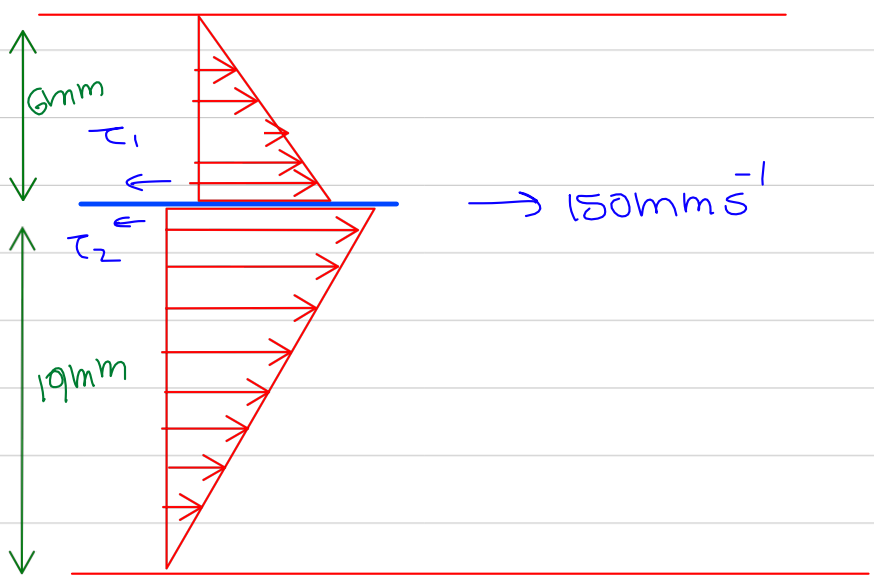
Bulk modulus $= k$

$k \uparrow \rightarrow$ incompressible (water)

$k \downarrow \rightarrow$ compressible (gas)

$\mu = 0.7$ Pas

Q1) Two parallel walls spaced at 25 mm is filled with a liquid of dynamic viscosity 0.7 Pa s. A thin flat plate of 250 mm x 250 mm is towed into this space at a velocity of 150 mm/s at a distance of 6 mm from one wall keeping the movement parallel to the walls. Assuming linear velocity variations between the plate and the walls, determine the shear force exerted by the liquid on the plate.



$V = 150 \ mm s^{-1}$

$A = 250 \times 250 \ mm^2$

$= 0.0625 \ m^2$

$\tau = \mu \frac{du}{dy}$

μ = dynamic viscosity

$\frac{du}{dy}$ = velocity change with gap distance

τ = Shear Force

$\tau = \mu \frac{du}{dy}$

$\frac{du}{dy} = \left(\frac{150 - 0}{6}\right) \frac{mm s^{-1}}{mm}$

$\left(\frac{du}{dy}\right)_1 = 25$

$\left(\frac{du}{dy}\right)_2 = \left(\frac{150 - 0}{19}\right)$

$\left(\frac{du}{dy}\right)_2 = 7.89$

$\mu = 0.7$ Pa.S

$\tau_1 = 0.7 \times 25$

$= 17.5$

$F_1 = 17.5 \times 0.0625$

$F_1 = 1.1 \ N$

We know that $F = PA$

therefore $F_1 = \tau_1 \times A$

$F_2 = \tau_2 \times A$

$\tau_2 = 0.7 \times 7.89$

$= 5.523$

$F_2 = 5.523 \times 0.0625$

$F_2 = 0.35 \ N$

Total Shear Force $= F_1 + F_2$

$F_1 + F_2 = 0.35 N + 1.1 N$

$F_1 + F_2 = 1.45 \ N$

Properties of water and air

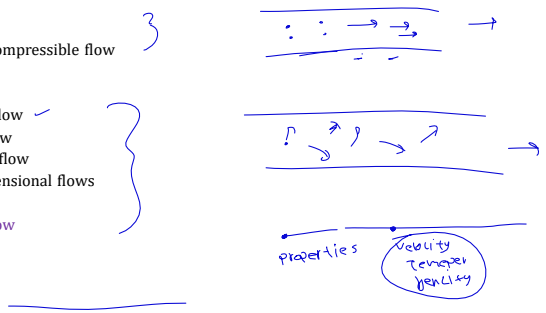
Water	
Mass density (ρ)	10^3 (kg/m^3)
Viscosity at 20^{0}C (μ)	10^{-3} (Pa.s)
Kinematic Viscosity at 20^{0}C (υ)	10^{-6} (m^2/s)
Air	
Mass density (ρ)	1.3 (kg/m^3)
Viscosity at s.t.p. (μ) (1/60 th that of water)	1.7(10^{-5}) Pa.s)
Kinematic viscosity at s.t.p. (υ) (13 times that of water)	1.3(10^{-5}) (m^2/s)
Gas constant (R=p/ρT)	287 (J/kg.K)
Specific heat capacity at constant pressure (C_p)	1000 (J/kg.K)
Specific heat capacity at constant volume (C_v)	715 (J/kg.K)
Ratio of specific heat capacities (γ)	1.4
Vapour pressure at 20^{0}C	2340 (N/m^2)
Bulk modulus of elasticity at 30^{0}C	2.25 (10^6) kN/m^2

Classification of fluid flows

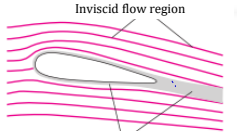
- In practice, there is a wide variety of fluid flows
- It is usually convenient to classify them on the basis of some common characteristics

Based on fluid properties
- Viscous versus Inviscid
- Compressible versus incompressible flow

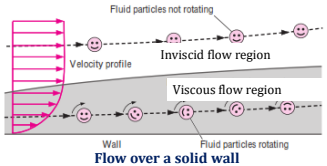
Based on flow properties
- Laminar versus turbulent flow
- Steady versus unsteady flow
- Uniform and non-uniform flow
- One-, Two-, or Three- dimensional flows
- Natural versus forced flow
- Internal versus external flow



Viscous versus inviscid regions of flow



Flow across an airfoil



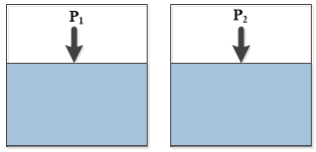
Flow over a solid wall

- When fluid layers move relative to each other, a friction force develops between them. The slower layer tries to slow down the faster layer.
- This internal resistance to flow is quantified by the fluid property *viscosity*
- Viscosity is caused by cohesive forces between the molecules in liquids and by molecular collisions in gases
- All fluid flows are viscous

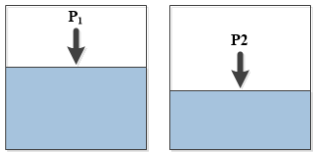
Compressible versus incompressible

Incompressible: density of fluid is constant

Compressible: density varies



Incompressible

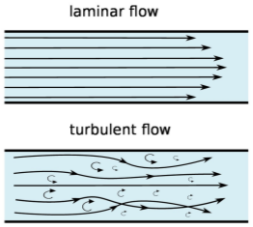


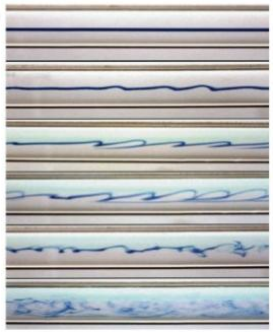
Compressible

Laminar and turbulent flow

- The flow in laminations (layers) is termed as laminar flow
- Fluid particles mix each other in turbulent flow
- Reynolds number is used to differentiate laminar and turbulent flow

$$Re = \frac{\rho V d}{\mu}$$





Transition from laminar to turbulent flow

Q3)A fluid having a density of 900 kg/m^3 and a kinematic viscosity of 4.2 × 10^{-4} m^2/s flows in a 50 mm diameter pipe at a rate of 2.5 l/s.

(i)Calculate the Reynolds number for the flow.

(i)What is the type of flow?

$$Re = \frac{\rho V d}{\mu}$$

Steady and Unsteady flow

Steady flow:

Flow properties (velocity, density, pressure etc.,) do not vary with time at a particular point

Unsteady flow:

Flow properties (velocity, density, pressure etc.,) do vary with time at a particular point

Uniform and Non-uniform flow

Uniform flow:

Flow conditions remains constant irrespective of the position

Non-uniform flow:

Flow conditions vary from one point to another

Q3)A fluid having a density of $900\ kg/m^3$ and a kinematic viscosity of $4.2 \times 10^{-4}\ m^2/s$ flows in a 50 mm diameter pipe at a rate of $2.5\ l/s$.

(i)Calculate the Reynolds number for the flow.

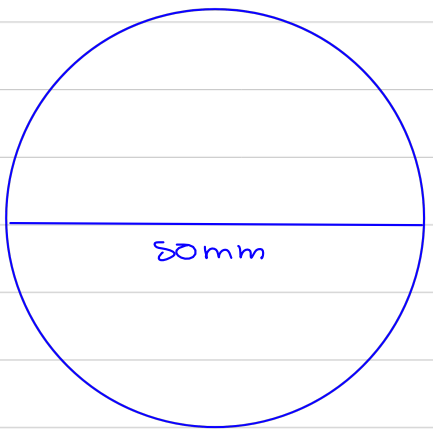
(i)What is the type of flow?

$Re = \frac{\rho V d}{\mu}$

$Re = \frac{Vd}{(\mu/\rho)} = \frac{Vd}{\nu}$

Reynold's Number =

$Re = \frac{\rho V d}{\mu}$ ← using dynamic viscosity



$$Re = \frac{1 \times \frac{2.5 \times 10^{-3} \times 4}{3.14 \times 50^2 \times 10^{-6}}\ ms^{-1} \times 50 \times 10^{-3}\ m}{4.2 \times 10^{-4}\ m^2 s^{-1}}$$

$A = \frac{\pi d^2}{4}$

$= \frac{3.14 \times 50^2 \times 10^{-6}}{4}$

$Re = 151.653$

$Re = \left(\frac{\rho V d}{\mu}\right)$ ν = kinematic viscosity

$= \left(\frac{Vd}{(\mu/\rho)}\right)$ μ = dynamic viscosity

$Re = \left(\frac{Vd}{\nu}\right)$ $\nu = (\mu/\rho)$

The formula for flow inside a pipe is:

$$Re = \frac{\rho V D}{\mu}$$

Where:

- ρ = Density of the fluid
- V = Velocity of the fluid
- D = Diameter of the pipe
- μ = Dynamic viscosity

How to classify the flow based on Re:

For standard flow inside a closed pipe, we use these widely accepted thresholds:

- **Laminar Flow ($Re \leq 2300$):** Viscous forces dominate. The fluid moves in smooth, parallel layers.
- **Transitional Flow ($2300 < Re < 4000$):** The flow is unstable, switching back and forth between smooth and chaotic.
- **Turbulent Flow ($Re \geq 4000$):** Inertial forces dominate. The flow is highly chaotic, with mixing, swirls, and eddies.

One, Two, and Three dimensional(1D, 2D, and 3D) flows

- Generally, all fluid flows are 3D, fluid properties vary in all directions
- In many cases, significant changes in two directions or in one directions. Thereby, 3D fluid flow can be simplified to 1D or 2D

1D: Flow parameters vary significantly only in one direction

2D: Flow parameters vary along the direction of flow and in one direction right angle to it

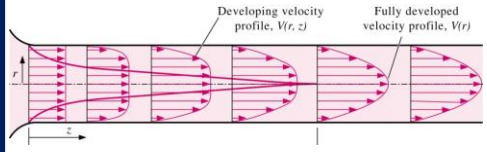
3D: Flow parameters vary significantly in all directions

Natural versus forced flow

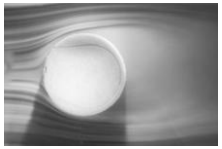
- Based on how the fluid motion is initiated
- F**orced flow, a fluid is forced to flow over a surface or** in a pipe by external means such as a pump or a fan
- **Natural flows, any** fluid motion is due to natural means

Internal versus external

- Depending on whether the fluid is forced to flow in a confined channel or over a surface.



Internal flow
Fluid is completely bounded by solid surfaces
- Flow in a pipe



External flow
Flow of an unbounded fluid over a surface
- Flow over cylinder
- Open channel flow

Dimensions and Units

- Any physical quantity can be characterized by **dimensions. The magnitudes** assigned to the dimensions are called **units.**
- Two commonly use unit systems; English systems, **metric International System (SI)**

Primary dimensions and their units in SI

Dimension	Unit
Length	meter (m)
Mass	kilogram (kg)
Time	second (s)
Temperature	kelvin (K)
Electric current	ampere (A)
Amount of light	candela(cd)
Amount of matter	mole(mol)

Some derived units

Quantity	Unit
Force	Newton (N or $kgms^{-2}$)
Pressure and stress	Pascal (Pa or Nm^{-2})
Work, energy	Joule (J or Nm)

- To avoid inconveniently large or small numbers, prefixes may be put in front of the unit names

Multiple	Prefix	Multiple	Prefix	Multiple	Prefix
10^{12}	Tera (T)	10^{2}	Hecto (h)	10^{-3}	milli (m)
10^{9}	Giga (G)	10^{1}	Deka (d)	10^{-6}	micro (μ)
10^{6}	Mega (M)	10^{-1}	deci (d)	10^{-9}	nano (n)
10^{3}	Kilo (k)	10^{-2}	centi (c)	10^{-12}	pico (p)

Dimensional homogeneity

- Equations must be dimensionally homogeneous.

  Every term in an equation must have the same unit
- The notation for the four primary dimensions is as follows: mass [M], length [L], time [T] and temperature [Θ].

Volumetric flow rate Q through a hole, distance h from the liquid surface and of diameter D on a side of a tank is given by

$$\underbrace{Q}_{\frac{L^3}{T}} = 0.68 \underbrace{D^2}_{L^2} \underbrace{\sqrt{gh}}_{\left(\frac{L}{T^2} L\right)^{1/2}}$$

$$\frac{L^3}{T}$$

Accuracy, Precision, and Significant digits

Engineers must be aware of three principals that govern the proper use of numbers.

Accuracy error : Value of one reading minus the true value. Closeness of the average reading to the true value. Generally associated with repeatable, fixed errors.

Precision error : Value of one reading minus the average of readings. Is a measure of the fineness of resolution and repeatability of the instrument. Generally associated with random errors.

Significant digits : Digits that are relevant and meaningful. When performing calculations, the final result is only as precise as the least precise parameter in the problem. When the number of significant digits is unknown, the accepted standard is 3. Use 3 in all homework and exams.